

MANGESH RAJENDRA SAVANT

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Software Engineer

Systems-focused software engineer at VIT Pune (GPA 9.55) with hands-on experience in distributed systems, compiler design, and memory management. Built production-grade Go systems including a Raft-based key-value store and a compiler frontend.

EDUCATION

Vishwakarma Institute of Technology, Pune

Aug 2024 – Present

B.Tech – Artificial Intelligence | **GPA: 9.55**

Relevant Coursework: Operating Systems, Computer Networks, DBMS, Data Structures & Algorithms

KRT Arts, BH Commerce & AM Science College, Nashik

Jun 2024

12th HSC Board | **87.67%**

Seth Lalji Narayanji Sarvajanik Vidyalaya, Jalgaon

Jun 2022

10th SSC Board | **93.80%**

PROJECTS

Distributed Key-Value Store with Raft Consensus

2025

- Designed and implemented a fault-tolerant distributed key-value store in Go using the Raft consensus protocol, supporting leader election, log replication, and cluster membership changes.
- Built a linearizable read/write API with gRPC, achieving strong consistency guarantees across a 5-node cluster under simulated network partitions and node failures.
- Engineered a compaction and snapshotting mechanism to bound log growth, reducing memory footprint by 60% in long-running clusters during stress tests.

Technologies: Go, Raft Consensus, gRPC, Distributed Systems, Concurrency, Data Structures

AST Generator (Compiler Frontend)

Jan 2026

- Developed a production-grade Abstract Syntax Tree Generator in Go capable of parsing a custom programming language and producing structured syntax trees for program analysis.
- Implemented tokenization, syntax validation, operator precedence parsing (Pratt parser), and compiler-style error reporting with precise line and column tracking.
- Designed an extensible AST and Visitor Pattern framework to support future phases including semantic analysis and LLVM IR code generation.

Technologies: Go, Compiler Design, AST, Parsing Algorithms, Visitor Pattern, JSON, Data Structures

Mark and Sweep Garbage Collection Algorithm

Jan 2025

- Implemented a Mark and Sweep Garbage Collector in Python featuring object tracking, root set management, and automatic reclamation of unreachable memory objects.
- Developed graph-based reachability analysis to correctly handle circular references, simulating core techniques used in production runtime memory managers.
- Created visualization and statistical monitoring modules for analyzing GC cycle performance, pause times, and object lifecycle patterns.

Technologies: Python, Garbage Collection, Graph Algorithms, OOP, Graphviz

BHUJAL-NETRA – Real-Time Groundwater Evaluation Platform

2025

- Built a full-stack monitoring platform covering 5,260+ DWLR stations across India, delivering four user-facing products: Flutter mobile app, research web portal, policymaker dashboard, and GIS map layer.
- Designed the system architecture to integrate real-time sensor data ingestion, multi-model forecasting (LSTM + Pastas), and a multilingual chatbot into a single cohesive platform.

- Reached Local Finals at Smart India Hackathon 2025; handled data at national scale with Firebase and Google Maps API integrations.

Technologies: *Python, Flutter, Firebase, LSTM, Pastas, Google Maps API, Matplotlib*

Malware Detection System

Jan 2026

- Built an end-to-end ML pipeline for classifying malicious versus benign executables using behavioral and feature-based analysis with Scikit-learn.
- Engineered preprocessing, feature selection, and classification stages; optimized for real-time detection of suspicious file patterns and activities.

Technologies: *Python, Scikit-learn, Machine Learning, Cybersecurity, Data Analysis*

TECHNICAL SKILLS

Domain	Tools & Technologies
Systems & Compiler Design	Go, AST, Parsing Algorithms, Visitor Pattern, LLVM IR, Tokenization
Programming Languages	Python, Go, C, C++, Java, JavaScript
Full Stack Development	React, Flutter, Streamlit, HTML, CSS, REST APIs, Firebase
Data Structures & Algorithms	Graph Algorithms, OOP, Memory Management, Recursion
DevOps & Tools	Git, Docker, Linux/Unix, MySQL, Android Studio, Graphviz, JSON, ESP32/IoT

EXTRA-CURRICULAR ACHIEVEMENTS

Theatre – Multiple Awards (2025)

- Firodiya Karandak 2025: 1st Prize – 3D Miniature
- Sarpotdar Karandak 2025: Best Actor – 3rd Prize
- Vinodottam Karandak 2025: Best Poster Making – 1st Prize
- Crescendo Natyasamrat: 1st Prize
- Active participant in 7+ theatre productions as Actor, Technical Coordinator, Sound Engineer, and Backstage Crew — demonstrating strong communication, teamwork, and leadership under pressure.